Eduardo Pane

Machine Learning Engineer — Computer Vision & Embedded AI

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Experience

Computer Vision Engineer, Consultant (Part-time) — FlexSight *Mar 2026 – Present*

- Consult on open or stalled industrial computer vision projects: analyze blockers, propose alternative approaches and prototype solutions
- Joined after a thesis in collaboration with the company on automated detection in industrial blueprints (see Education)

CTO — HERALD, Visual Localization & Counter-UAS *May 2026 – Sep 2026*

- Led technical development of a GPS-denied system estimating a drone's position by matching its video feed against satellite imagery (132-tile mosaic, 3072×2816 px, 0.73 m/px)
- Designed a coarse-to-fine search cutting per-frame matching from 9,628 windows (~12 min) to 132 global windows (~13 s) and ~19 local windows with motion momentum (~1–2 s); ~5 s/frame average on simple scenarios
- SuperPoint / SuperGlue matching: 70–150 inliers in open areas vs. 3–5 in urban zones; validated on MovingDrone sequences up to 104 frames, with pySLAM visual odometry tracking 15/15 frames

Education

M.Sc. Computer Engineering — Artificial Intelligence, University of Padova *Sep 2026 – Present*

B.Sc. Computer Engineering, University of Padova *Sep 2023 – Jul 2026*

- Final Grade: 108/110
- Thesis (with FlexSight): synthetic-data-driven pipeline detecting GD&T annotations in industrial blueprints. Built a generator with 12 layout variants and exact bounding-box reprojection, plus class-adaptive augmentation (up to 10x for rare classes); reached 0.934 mAP@50 on the dimension detector and 0.852 mAP@50 on the symbol detector

Projects

LIMEN — AI-Powered Panic Attack Prevention Wearable *2026*

- 2-layer LSTM on HRV, EDA and temperature reaching 97% validation accuracy; 1st place at a hardware hackathon
- 8-bit quantized and deployed on nRF5340 for real-time on-device inference with zero cloud dependency, with IMU-based activity filtering

Hollow Knight AI — Deep Reinforcement Learning Agent *In Development*

- DQN agent in PyTorch (10,000+ LOC) with experience replay and target networks, trained against the Nightmare King Grimm boss
- Real-time YOLOv8 vision pipeline at 30 FPS tracking player, boss animations and projectiles; custom reward shaping with 50+ tunable parameters

Multi-Agent Career & Opportunity Research System *2026*

- Multi-agent system in Python and CrewAI coordinating specialized agents that surface hackathons, fellowships and programs with verified application windows
- Outreach agents identify relevant professionals in a target field and draft personalized, context-aware messages

Reinforcement Learning Game AI — Personal Projects *2024 – 2026*

- Four agents built from scratch in Python and C++: tabular Q-Learning on Arduino Mega (8KB RAM), multi-agent DQN (Pong), YOLOv8 webcam AR (Flappy Bird), genetic algorithm with 6-layer network (Snake)

Technical Skills

Languages: Python, C/C++, Java, SQL, TypeScript
ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost
Computer Vision: OpenCV, YOLO, Object Detection, Synthetic Data Generation, Multi-Camera Systems
Embedded ML: TinyML, Model Quantization (int8), nRF5340, Edge AI deployment
Other: CrewAI, LLM Orchestration, Multi-Agent Systems, Git, Linux, Docker
Certifications: Deep Learning Specialization (DeepLearning.AI, 2026), Machine Learning Specialization (Stanford / Coursera, 2025)

Selected Awards & Fellowships

- Fellow — Kickstart Fellowship, Autumn 2026 Cohort
- 1st Place — Startup Generation Challenge, RelazionEXPO Verona (with Economics Network) — Oct 2026
- 1st Place — “I Nuovi Dispositivi del Futuro” Hackathon, Elevator Innovation Hub Vicenza (LIMEN; internship offer) — Mar 2026
- Aikido Security Prize — Most Secure App, Big Berlin Hack, Tech Europe (158 teams; contacted by Reonic) — Apr 2026
- 3rd / 27 Teams — Defense-Tech Hackathon, UAV Interceptor (200 participants) — May 2026
- Ambassador — Hackathon Hub — Jun 2026 – Present
- Kaggle — Avg. Top 30% across 5 competitions (2,000–5,000+ participants)
- Italian Mathematical and Informatics Olympiads — Regional Finals Qualifier, 2 consecutive years (2020–2021)